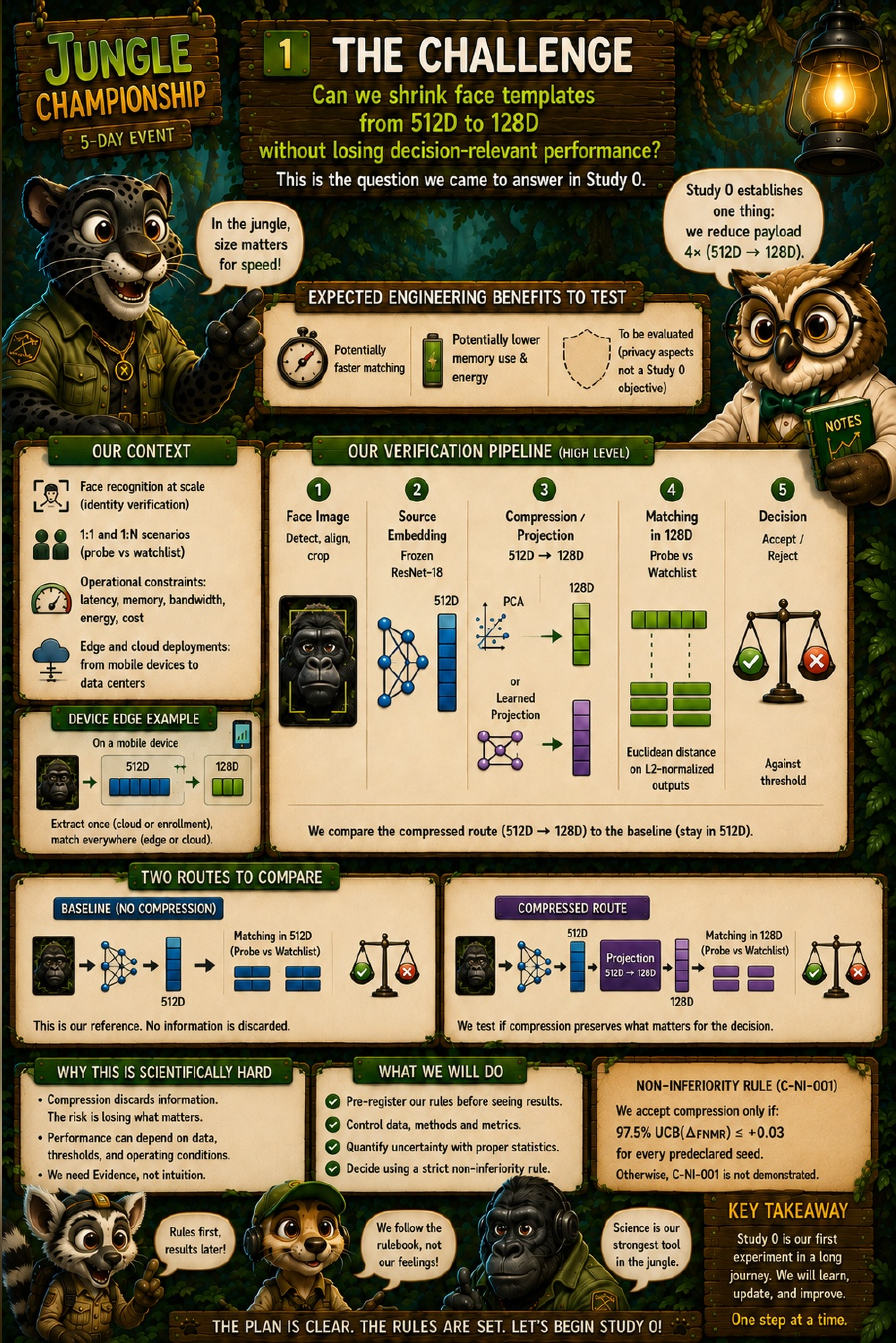

JUNGLE
CHAMPIONSHIP
A crazy product idea.
A serious scientific lesson.
THE JUNGLE CHAMPIONSHIP ARRIVES
The PM wants to propose a near 0 integration, edge live face watchlist search device.
BUT ENGINEERING TEAMS PINPOINT
probable memory and response time challenges.
FIX MY PROPOSAL
Make it work at the edge.
POPULATION ALREADY ENROLLED
Face watchlist already stores 512-feature templates per portrait.
BATRICIA
PRODUCT MANAGER
JUNIOR
FIELD INTEGRATOR
MAURICE
GILBERT
SCIENTIST & STATISTICIAN
Study 0 — Siamese embedding compression at the edge


JUNGLE CHAMPIONSHIP
5-DAY EVENT
1 THE CHALLENGE
Can we shrink face templates
from 512D to 128D
without losing decision-relevant performance?
This is the question we came to answer in Study 0.
In the jungle, size matters for speed!
Study 0 establishes one thing: we reduce payload 4× (512D → 128D).
EXPECTED ENGINEERING BENEFITS TO TEST
Potentially faster matching
Potentially lower memory use & energy
To be evaluated (privacy aspects not a Study 0 objective)
NOTES
OUR CONTEXT
Face recognition at scale (identity verification)
1:1 and 1:N scenarios (probe vs watchlist)
Operational constraints: latency, memory, bandwidth, energy, cost
Edge and cloud deployments: from mobile devices to data centers
DEVICE EDGE EXAMPLE
On a mobile device
512D
128D
Extract once (cloud or enrollment), match everywhere (edge or cloud).
OUR VERIFICATION PIPELINE (HIGH LEVEL)
1 Face Image
Detect, align, crop
2 Source Embedding
Frozen ResNet-18
512D
3 Compression / Projection
512D → 128D
PCA
128D
or
Learned Projection
4 Matching in 128D
Probe vs Watchlist
Euclidean distance on L2-normalized outputs
5 Decision
Accept / Reject
Against threshold
We compare the compressed route (512D → 128D) to the baseline (stay in 512D).
TWO ROUTES TO COMPARE
BASELINE (NO COMPRESSION)
Matching in 512D (Probe vs Watchlist)
512D
This is our reference. No information is discarded.
COMPRESSED ROUTE
512D
Projection 512D → 128D
Matching in 128D (Probe vs Watchlist)
128D
We test if compression preserves what matters for the decision.
WHY THIS IS SCIENTIFICALLY HARD
• Compression discards information. The risk is losing what matters.
• Performance can depend on data, thresholds, and operating conditions.
• We need Evidence, not intuition.
WHAT WE WILL DO
Pre-register our rules before seeing results.
Control data, methods and metrics.
Quantify uncertainty with proper statistics.
Decide using a strict non-inferiority rule.
NON-INFERIORITY RULE (C-NI-001)
We accept compression only if:
97.5% UCB(ΔFNMR) ≤ +0.03
for every predeclared seed.
Otherwise, C-NI-001 is not demonstrated.
Rules first, results later!
We follow the rulebook, not our feelings!
Science is our strongest tool in the jungle.
KEY TAKEAWAY
Study 0 is our first experiment in a long journey. We will learn, update, and improve.
One step at a time.
THE PLAN IS CLEAR. THE RULES ARE SET. LET'S BEGIN STUDY 0!

JUNGLE CHAMPIONSHIP
5-DAY EVENT

2 STUDY 0: OUR RULEBOOK

Before we run, we agree on the rules.
No rules, no science!

A good explorer checks the map before leaving!

Our rulebook is our contract with future scientists (and with ourselves).

OUR MISSION

Test a claim in biometric verification:

"We can compress faces from 512D to 128D without losing decision-relevant performance."

Decision metric: $\Delta = \text{metric}_{proj} - \text{metric}_{baseline}$

Decision rule: 97.5% UCB(ΔFNMR) ≤ +0.03 for every predeclared seed.

RULES

THE RULES WE FIXED BEFORE RUNNING

1. HYPOTHESIS & CLAIM(S)
- What we claim
- What we do not claim
- Success criterion (C-NI-001)

2. DATA & REPRESENTATIVITY
- LFW (cleaned)
- Min 10 identities × 2 imgs
- ≥ 500 impostor pairs
- Why this population

3. METHODS & METRICS
- Embedding extractor (frozen)
- Projection / compression
- Matching: Euclidean distance on L2-normalized outputs
- Metrics: FMR@FMR*, FNMR@FMR*, and Δ

4. ESTIMATOR & UNCERTAINTY
- Bootstrap plan (subject-level)
- CI type (e.g., percentile)
- Seed strategy

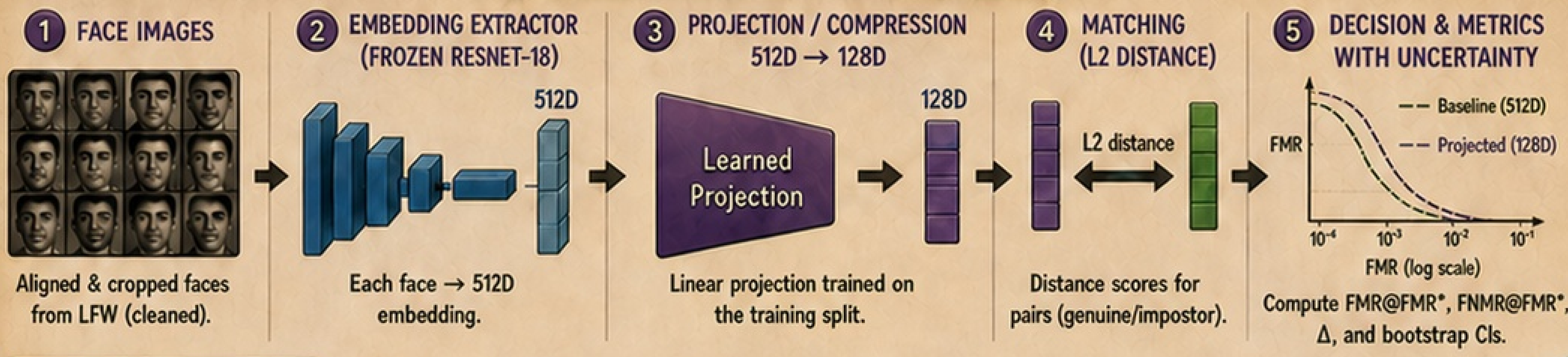
5. DECISION RULE (C-NI-001)
- Non-inferiority margin ε
- 97.5% UCB(ΔFNMR) ≤ +0.03 for every predeclared seed
- Predefined conclusion

6. REPRODUCIBILITY & OPENING
- Code & notebooks
- Protocol versioning
- Results disclosure

★ We write everything down BEFORE we look at results. This protects us from wishful thinking!

Future us will thank us!

STUDY 0 DESIGN AT A GLANCE



SPLITS (PRE-REGISTERED)

- Train (projection only) ~60% of identities.
- Validation (tune λ, keep blind) ~20% of identities
- Test (final evaluation) ~20% of identities
- No identity overlap across splits!

Great scientists are great rule-followers!

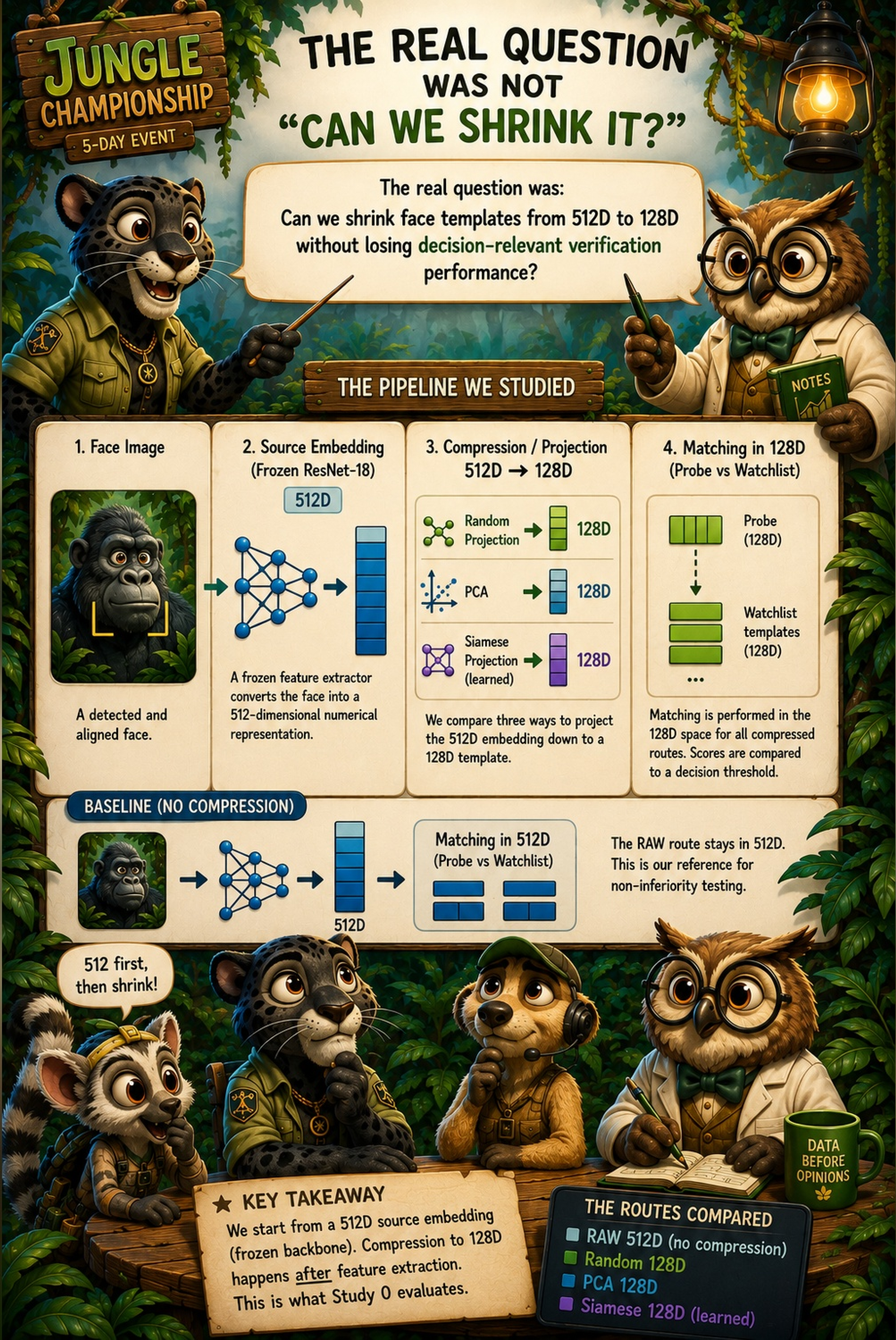
WHY THESE RULES MATTER

NEXT UP:

With our rulebook in place, we're ready to package everything—code, data, and protocol—into our Replay Bundle.

REPLAY BUNDLE

JUNGLE CHAMPIONSHIP
5-DAY EVENT
3. REPLAY BUNDLE
PROTOCOL v0.2.2 – FROZEN
We turned the experiment into a replayable, versioned bundle.
Same steps. Same data. Same environment.
Anyone can replay and get the same result.
We freeze what matters, so science never depends on memory.
REPLAY CHECKLIST
Data
Code
Environment
Results
Logs
WHAT'S INSIDE THE REPLAY BUNDLE
A. PROTOCOL v0.2.2 (FROZEN)
The exact recipe.
PROTOCOL v0.2.2 (FROZEN)
Data sources & splits
Models & hyperparameters
Training procedure
Projection / metric settings
Evaluation rules & estimands
Statistical analysis plan
No changes allowed without a new version.
B. CODE & NOTEBOOKS
The exact implementation.
</>
Training scripts
Projection & metric code
Evaluation & bootstrap code
Analysis notebook
Requirements.txt
README with instructions
Run once. Run twice. Same code → same result.
C. RESULTS & LOGS
The evidence.
LOGS .jsonl
Summary metrics (JSON/CSV)
Confidence intervals
Plots & figures
Bootstrap distributions
Execution logs (progress.jsonl)
Performance (time / memory)
Transparent. Traceable. Auditable.
D. ENVIRONMENT SNAPSHOT
The exact environment.
Python version
Package versions (pip/conda export)
OS / GPU info
Deterministic seeds
Hardware profile
Environment manifest archived
We capture the full context so the result can be replayed exactly.
WHY THIS MATTERS
Reproducibility: anyone can replay.
Integrity: nothing can change by accident.
Auditability: everything is traceable.
Credibility: results we can trust.
OUR COMMITMENT
We don't just run experiments.
We preserve them.
So the result belongs to the science, not to one moment in time.
READY TO REPLAY
Everything is in the bundle.
Unzip. Follow the instructions.
You'll get the same result.
REPLAY BUNDLE
RUN
SAME RESULT


JUNGLE CHAMPIONSHIP
5-DAY EVENT
THE REAL QUESTION WAS NOT "CAN WE SHRINK IT?"
The real question was: Can we shrink face templates from 512D to 128D without losing decision-relevant verification performance?
THE PIPELINE WE STUDIED
1. Face Image
A detected and aligned face.
2. Source Embedding (Frozen ResNet-18)
512D
A frozen feature extractor converts the face into a 512-dimensional numerical representation.
3. Compression / Projection 512D → 128D
Random Projection → 128D
PCA → 128D
Siamese Projection (learned) → 128D
We compare three ways to project the 512D embedding down to a 128D template.
4. Matching in 128D (Probe vs Watchlist)
Probe (128D)
Watchlist templates (128D)
...
Matching is performed in the 128D space for all compressed routes. Scores are compared to a decision threshold.
BASELINE (NO COMPRESSION)
512D
Matching in 512D (Probe vs Watchlist)
The RAW route stays in 512D. This is our reference for non-inferiority testing.
NOTES
512 first, then shrink!
KEY TAKEAWAY
We start from a 512D source embedding (frozen backbone). Compression to 128D happens after feature extraction. This is what Study 0 evaluates.
DATA BEFORE OPINIONS
THE ROUTES COMPARED
RAW 512D (no compression)
Random 128D
PCA 128D
Siamese 128D (learned)

JUNGLE CHAMPIONSHIP
5-DAY EVENT
Study 0, in simple words
We did not ask which method looked coolest. We asked which compressed route preserved the operating decision we care about.
1 Start from a frozen source embedding extractor
Frozen ResNet-18
2 Create a 512D source embedding
512D
3 Compare four routes
RAW 512D (no compression)
Random 128D (compressed)
PCA 128D (compressed)
Siamese 128D (compressed)
Routes 2–4 compress from 512D → 128D
4 Judge at false-match rate = 0.01
False matches (should be rare)
Threshold
5 Allow at most +0.03 FNMR loss
RAW 512D
Baseline FNMR
Compressed route
Allowed up to +0.03
Non-inferiority margin (+0.03 FNMR)
6 Every predeclared seed must pass
11
29
47
71
101
All 5 seeds must satisfy the rule for the route to pass.
NOTES
DATA BEFORE OPINIONS
So all 5 seeds must pass?!
Why so strict?
In biometrics, decisions are trade-offs. Predeclaring the rule prevents us from moving the goalposts after seeing results.
RULE CHECK
FMR = 0.01
+0.03 FNMR
All 5 seeds
GLOSSARY
FMR: false matches accepted
FNMR: true matches rejected
Non-inferiority: not worse beyond a predefined margin
Seed: a predeclared random initialization / repeat setting
Jungle Championship • Study 0 rulebook

JUNGLE CHAMPIONSHIP
5-DAY EVENT

6. THE PROBLEM APPEARED LATER

The result looked precise. Then Gilbert checked the uncertainty.

During a later methodological audit of Study 0, we found a hidden problem.

WHAT THE FIRST ANALYSIS DID

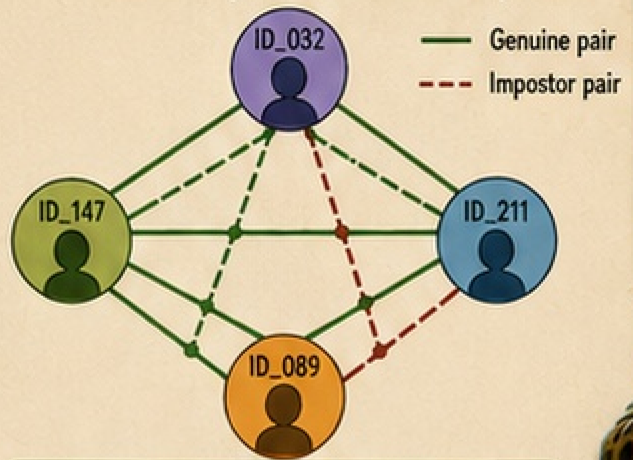
Bootstrap pair rows independently
(rows = treated as independent observations)

Pair ID	Subject A	Subject B	Label
1	ID_032	ID_147	Genuine
2	ID_032	ID_211	Genuine
3	ID_032	ID_211	Genuine
4	ID_147	ID_211	Impostor
5	ID_147	ID_032	Genuine
6	ID_089	ID_211	Impostor
...	...	...	...

Treats each row as if it were independent. But many rows share the same identities. That's not independent evidence.

WHAT THE DATA REALLY CONTAINED

Pairs share subjects → evidence is dependent



Each identity participates in multiple pairs. These pairs are correlated, not independent.

It looks precise!

Great! Positive with a narrow CI!

ORIGINAL RESULT (PAIR-LEVEL BOOTSTRAP)

Mean Δ = +0.0228

Shall we celebrate?

WHY THIS MATTERS

Repeated subjects → Correlated pair rows → Uncertainty underestimated → Confidence intervals too narrow → Near the non-inferiority limit, this can lead to a wrong scientific decision.

After correction, subject-level intervals were about **1.5× wider** than the original pair-level intervals.

The bug was not necessarily in the point estimate. It was in how certain we thought we were.

JUNGLE CHAMPIONSHIP
5-DAY EVENT

7. THE FIX WAS MORE THAN A BUG FIX

We corrected the estimator and protected the whole experiment.

EVIDENCE PROTECTED

1. FROZEN INPUTS (UNCHANGED)

Same data, same source embeddings, same train/val/test splits. Nothing changed.

2. CORRECTED ESTIMATOR (SUBJECT-BOOTSTRAP)

We resample identities (subjects), not pair rows. All pairs of the selected identities are included together in each resample.

3. REPLAY BUNDLE (PROTOCOL v0.2.2 – FROZEN)

REPLAY BUNDLE

- ✓ Protocol v0.2.2 (frozen)
- ✓ Code & notebooks
- ✓ Results & logs
- ✓ Environment snapshot

4. INDEPENDENT VERIFICATION

Maurice reran the process independently using only the replay bundle. Results matched exactly.

DISCIPLINE BUILDS TRUST

WHAT CHANGED

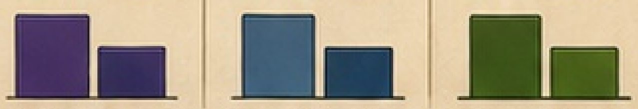
Before (wrong unit):
Resample pair rows
→ underestimates uncertainty ✗

After (correct unit):
Resample identities (subjects)
→ correct uncertainty ✓

THE IMPACT

Subject-level intervals were materially wider.

Random: ~1.57× wider | PCA: ~1.52× wider | Siamese: ~1.54× wider



Wider uncertainty. Same direction. Stronger evidence.

SCIENTIFIC DIRECTION UNCHANGED

The correction did not rescue or reverse the claim. It showed that the original analysis was too confident.

C-NI-001: NOT DEMONSTRATED

Frozen downstream rule:
97.5% UCB of ΔFNMR ≤ +0.03
for every predeclared seed.

This was not just a patch. It was a correction of the scientific record.

✓ Accurate uncertainty ✓ Reproducible evidence ✓ Decisions we can defend

JUNGLE CHAMPIONSHIP
5-DAY EVENT

8. THE RESULT

Final decision under the preregistered rule.

Here are the final results under our preregistered rule. 0 of 5 seeds passed for each route.

PREREGISTERED RULE (DECISION RULE)

For every predeclared seed:

97.5% UCB(ΔFNMR) ≤ +0.03

Each route failed the rule. None demonstrated non-inferiority in Study 0.

Route	Seeds passing	97.5% UCB range
Random 128D	0/5	0.112649–0.151768
PCA 128D	0/5	0.125556–0.133606
Siamese 128D	0/5	0.176584–0.189156

HONEST DATA STRONG DECISIONS

DECISION

C-NI-001: NOT DEMONSTRATED

Under the preregistered rule, none of the three 128D routes demonstrated non-inferiority in Study 0, because 0 of 5 predeclared seeds passed for each route.

IMPORTANT CLARIFICATION

This does not prove that 128D compression is generally inferior.

Study 0 asked a specific, preregistered question under a strict rule. We need more evidence before drawing broader conclusions.

WHAT THIS MEANS

NO FALSE VICTORY
We avoided a potential false claim that could have misled the field.

REPRODUCIBLE EVIDENCE
Anyone repeating the study with the same protocol and data will get the same answer.

SCIENTIFIC INTEGRITY
We followed our pre-registered plan and let the rule—not hope—decide.

DECISIONS WE CAN DEFEND
We can defend this result in a review, an audit, or a real-world decision.

NEXT STEPS
We need more informative evidence (better data, other metrics, other projections) to move the needle.

EVIDENCE FIRST

BOTTOM LINE

We followed the science and the rule.
No route passed. Now we gather better evidence and keep moving forward—together.

NEXT ON THE TRAIL

- Study 1: Better data
- Study 2: Other metrics
- Study 3: Task-based validation

JUNGLE CHAMPIONSHIP

5-DAY EVENT

9. THE LESSONS WE'LL CARRY FORWARD

Science is not about being right. It's about being trustworthy.

This wasn't just one mistake. It' was a valuable lesson in how we do science.

WHAT WE LEARNED

1 DEPENDENCE MATTERS

Pairs that share the same subjects are correlated.

Treating them as independent can give a false sense of precision.

2 UNCERTAINTY PROTECT SCIENCE

A frozen, versioned protocol and replay bundle allow anyone to reach the same answer.

Reproducibility is our safety net.

REPLAY BUNDLE

3 CHECK THE UNIT OF EVIDENCE

Always ask: "What is the true independent unit in my data?"

Then design the estimator around it.

4 INTEGRITY OVER EGO

We would rather slow down to get it right than move fast and be wrong.

Correcting ourselves builds trust.

THE IMPACT ON STUDY 0

FROZEN RULE

97.5% UCB(ΔFNMR) ≤ +0.03 for every seed

Random 128D:	0/5
PCA 128D:	0/5
Siamese 128D:	0/5

FINAL DECISION

C-NI-001: NOT DEMONSTRATED

Under the preregistered rule, no route demonstrated non-inferiority in Study 0.

We corrected the uncertainty estimator, replayed the frozen analysis, and let the preregistered rule decide.

Now we know exactly how to level up our experiments!

BETTER DATA

More identities, more diversity, better impostor modeling.

BETTER METRICS

Use metrics that matter operationally (FMR, FNMR, CIIR) at scale.

BETTER METHODS

Explore metric learning, hard negative mining, and alternative losses.

BETTER EVIDENCE

Larger benchmarks, multiple sites, transparent reporting.

BETTER DECISIONS

Decisions based on credible evidence, not wishful thinking.

OUR COMMITMENTS

TRANSPARENCY

We share protocols, data dictionaries, and results — good and bad.

REPRODUCIBILITY

Any expert can replay our work and get the same answer.

RIGOR

We challenge our assumptions and stress our methods.

CONTINUOUS IMPROVEMENT

We learn, adapt, and iterate.

SCIENTIFIC INTEGRITY

We do the right thing, even when no one is watching.

Trust is earned one careful step at a time.

NEXT STEPS

- Better data
- Face-specific backbone
- Better stress/qualification datasets
- Matched controls
- Progressive evidence

FINAL TAKEAWAY

We didn't win by finding a better number.
We won by earning the right to be believed.
That's how science advances.